

ATHINAI ELEFThERIOS VENIZELOS, Greece

WMO: 167161

Lat: 37.936N

Lon: 23.944E

Elev: 94

StdP: 100.20

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	2.0	-6.1	2.3	4.5	-4.1	2.7	4.8	14.3	10.2	12.6	10.0	3.2	280	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	35.0	20.3	33.2	20.0	32.2	19.7	22.2	30.1	21.5	29.7	21.0	29.4	6.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	25.9	18.9	13.9	25.5	18.0	13.1	25.2	65.5	30.2	63.2	29.6	61.1	29.1	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	10.9	9.9	-3.4	38.4	2.2	2.0	-4.9	39.9	-6.2	41.1	-7.5	42.2	-9.0	43.7		
			-4.2	23.5	2.1	0.7	-5.7	24.0	-7.0	24.5	-8.2	24.9	-9.7	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	9.0	9.7	11.9	15.0	19.4	24.1	26.9	27.2	22.9	18.2	14.2	10.5
	DBStd	6.97	3.33	3.39	2.72	2.63	2.64	2.88	1.97	1.89	2.50	2.69	3.10	3.27
	HDD10.0	152	59	42	13	1	0	0	0	0	0	0	4	33
	HDD18.3	1259	290	242	199	105	19	0	0	0	1	34	127	243
	CDD10.0	2876	27	34	72	149	292	422	524	532	388	256	131	49
	CDD18.3	942	0	0	0	3	52	173	266	273	139	32	4	1
	CDH23.3	10051	0	0	0	40	436	1848	3207	3261	1115	138	7	0
	CDH26.7	4041	0	0	0	3	85	723	1441	1447	329	12	1	0
Wind	WSAvg	3.9	3.8	4.0	3.9	3.7	3.4	3.8	4.7	5.0	4.0	3.9	3.5	3.6
Precipitation	PrecAvg	434	59	55	43	24	16	9	5	4	28	48	68	77
	PrecMax	632	141	138	99	57	70	24	20	33	116	98	192	192
	PrecMin	305	3	17	11	1	1	0	0	0	0	1	2	4
	PrecStd	85	33	38	24	18	16	8	6	8	31	32	51	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	19.2	21.2	26.9	31.0	37.1	37.8	36.1	33.8	28.1	24.1	20.2
		MCWB	12.9	13.1	13.4	14.8	17.9	20.1	20.5	20.4	19.8	18.5	17.3	15.4
	2%	DB	17.0	18.0	20.0	24.1	28.9	33.8	34.9	34.8	31.2	26.1	22.2	19.0
		MCWB	12.3	12.6	13.0	14.8	17.1	19.9	20.6	20.1	19.2	17.7	16.7	14.3
	5%	DB	15.9	16.8	18.8	22.8	27.2	32.0	33.2	33.2	30.0	24.9	21.0	17.9
		MCWB	11.8	12.2	12.4	14.1	16.4	19.4	20.0	20.1	18.7	17.4	16.3	13.5
	10%	DB	14.9	15.2	17.2	21.0	25.9	30.8	32.1	32.1	28.8	23.2	19.8	16.1
		MCWB	11.1	11.3	11.8	13.6	16.0	19.0	19.6	19.9	18.5	16.8	15.3	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	14.5	15.0	16.7	19.1	21.8	22.9	23.3	22.2	20.5	18.6	16.8
		MCDB	17.1	18.4	19.4	23.3	27.2	31.5	30.8	31.1	28.4	25.1	22.8	19.9
	2%	WB	13.1	13.4	13.9	15.7	18.2	21.0	22.0	22.5	21.2	19.6	17.6	15.0
		MCDB	15.9	16.7	18.2	21.7	26.0	30.6	30.4	30.3	27.1	24.0	21.4	18.0
	5%	WB	12.3	12.6	13.1	14.9	17.5	20.4	21.3	21.6	20.5	18.7	16.7	14.0
		MCDB	15.3	15.8	17.2	20.9	24.9	29.6	30.1	29.8	26.5	23.1	20.2	17.1
	10%	WB	11.2	11.7	12.4	14.1	16.9	19.7	20.8	21.0	19.7	17.8	15.8	12.8
		MCDB	14.3	14.9	16.6	19.8	23.8	28.6	29.9	29.6	26.1	22.1	19.1	15.6
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.8	8.6	10.1	11.0	10.8	10.0	9.5	9.5	8.5	8.2	7.8
		MCDBR	9.0	9.7	11.0	13.3	13.4	12.6	12.2	11.6	12.1	10.8	9.2	9.0
		MCWBR	6.1	6.2	6.2	6.7	5.5	4.7	4.7	4.7	5.3	5.4	5.6	6.0
	5% WB	MCDBR	7.9	8.7	9.7	11.6	11.7	11.5	11.1	10.5	10.7	9.7	8.2	8.0
		MCWBR	5.7	5.9	6.1	6.4	5.3	4.8	4.7	4.8	5.2	5.3	5.2	5.7
Clear-Sky Solar Irradiance	taub	0.350	0.374	0.409	0.450	0.432	0.411	0.414	0.420	0.419	0.395	0.375	0.360	
	taud	2.431	2.347	2.247	2.138	2.254	2.349	2.347	2.323	2.328	2.370	2.391	2.405	
	Ebn at Noon	823	846	848	832	854	869	863	850	829	812	783	780	
	Edh at Noon	86	105	128	151	137	124	124	124	117	102	89	83	
All-Sky Solar Radiation	RadAvg	1.98	2.76	4.16	5.49	6.57	7.48	7.59	6.76	5.20	3.54	2.35	1.72	
	RadStd	0.22	0.33	0.33	0.47	0.32	0.37	0.16	0.14	0.27	0.27	0.22	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-82	N/A	N/A

Nomenclature: See separate page